

EC508 Econometrics: Integrated Course Map

From MLR Foundations to IV, Panel, and Time Series

Class Summary from slides_andre

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Recap

Basic MLR Recap (Wooldridge)

Model:

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + u_i$$

Key assumptions:

- MLR.1 Linearity in parameters
- MLR.2 Random sampling
- MLR.3 No perfect collinearity
- MLR.4 Zero conditional mean: $E(u_i | X_i) = 0$
- MLR.5 Homoskedasticity (for usual SE formulas)

Why does it matter

- MLR.1–MLR.4 hold: unbiased/consistent
- MLR.5 also holds: efficient
- If normality is added: exact small-sample t and F inference

Typical failure map:

- Break MLR.4 $\Rightarrow$ bias/inconsistency (endogeneity)
- Break MLR.5 $\Rightarrow$ wrong classical SE/tests, inefficiency

1. Heteroskedasticity

Unit 1: Heteroskedasticity — What Changes?

$$\text{Var}(u_i | X_i) = \sigma^2 \quad (\text{homoskedasticity})$$

Changed setting:

$$\text{Var}(u_i | X_i) = \sigma_i^2$$

- Unbiased/consistent (if zero conditional mean holds)
- Lose efficiency (no longer BLUE)
- Lose inference: usual (non-robust) SE, t , F , confidence intervals are unreliable

Inference fix

Use heteroskedasticity-robust (Huber-White) standard errors for valid large-sample inference.

Robust Standard Errors

- Consistent under unknown heteroskedasticity forms
- $\hat{\beta}$ unchanged; only estimated uncertainty changes
- Robust SE can be larger or smaller; often larger in practice

- Breusch-Pagan (linear variance function idea)
- White test (more general nonlinear form using squares/interactions)

Logic:

- Regress $\hat{u}_i^2$ on functions of regressors
- Significant explanatory power in this auxiliary regression suggests heteroskedasticity

If variance form is known, weight:

$$\text{Var}(u_i | X_i) = \sigma^2 h_i$$

Use weighted least squares:

- Lower weight to high-variance observations
- Efficiency relative to OLS
- Bad variance specification can hurt

2. Data and Specification Issues

Other issues

- Functional form misspecification
- Proxy variables
- Measurement error (in y vs in x)
- Missing data and outliers

Functional Form

- RESET test
- Non-nested comparisons

Proxy Variables

- If error is simply random, you are good
- If not perfect, usually at least better
- In exam, decompose the variable into the proxy and the error

Missing Data and Selection

Cases:

- Missing at random: no bias, less precision
- Missing based on Y or unobservables: sample selection bias
- (In class: draw how can induce attenuation using scatter deleting high income people.)

Outliers and Influence

- OLS **squares** residuals, extreme points have large effect
- Report robustness unless a clear data error justifies exclusion

Units 1 and 2 Takeaway

- From heteroskedasticity: what breaks, how we solve
- From specification/data issues: present more problems that can threaten consistency besides the already known omitted variable approach

3. Time Series

Time Series vs Cross Section

- Ordering matters
- Can have seasonality and trends
- Random sampling hardly holds
- Lag structure, stationarity, trends, and serial dependence

Dynamics, interpretation and long-run effects

Static model:

$$y_t = \beta_0 + \beta_1 x_t + u_t$$

Finite distributed lag (FDL):

$$y_t = \delta_0 + \delta_1 x_t + \delta_2 x_{t-1} + \cdots + u_t$$

Assumptions: Time Series

Class assumptions TS.1–TS.5:

- TS.1 Linearity in parameters
- TS.2 No perfect collinearity
- TS.3 Strict exogeneity: $E(u_t | X) = 0$ (not X_t !) for all t
- TS.4 Homoskedasticity
- TS.5 No serial correlation in errors

What's new

- Endogeneity is a stronger condition
- OLS may remain unbiased under TS.1–TS.3
- But efficiency and classical inference rely on TS.4–TS.5

4. Time Series and Forecasting

AR(1):

$$y_t = \beta_0 + \beta_1 y_{t-1} + u_t$$

AR(p): p lags.

- Prediction vs. causality
- Lag choice on information criteria (AIC/BIC)

ADL Models and Granger Causality

ADL combines lags of y and predictors x :

$$y_t = \beta_0 + \sum_{j=1}^p \beta_j y_{t-j} + \sum_{m=1}^r \delta_m x_{t-m} + u_t$$

Granger causality test: (regretted calling it causality)

- Lagged x predicts y ?
- Predictive, not causal

Forecasting and Uncertainty

- Forecast error (out-of-sample vs in-sample)
- RMSE/RMSFE quantifies typical forecast mistake
- Build forecast intervals using variance of errors
- In-sample approximation
- Pseudo out-of-sample rolling/recursive evaluation

Non-Stationarity and Unit Roots

Random walk with drift:

$$y_t = \beta_0 + y_{t-1} + u_t$$

Implication:

- Stochastic trend, nonstationary
- Regressions in levels can be spurious
- Dickey-Fuller test for unit root

Unit Roots

- Standard asymptotics fail
- DF/ADF test:

$$\Delta y_t = \alpha_0 + \alpha_1 y_{t-1} + \sum_{j=1}^p \gamma_j \Delta y_{t-j} + v_t$$

- If $\alpha_1 = 0$: unit root, nonstationary. Need alternative critical values
- Differences are stationary (if $I(1)$): Δy_t

Takeaway

Where this fits

New data structure. Problems with sampling and spurious regressions.

5. Panel Data

- Repeated cross-section and panel data
- DGP:

$$y_{it} = \beta_0 + \beta_1 x_{it} + v_{it}, \quad v_{it} = a_i + u_{it}$$

- POLS: $\text{Cov}(x_{it}, a_i) = 0$ and $\text{Cov}(x_{it}, u_{it}) = 0$ for all t (if a_i correlated with x_{it} , OVB)

First Difference (FD) Estimator

Two-period illustration:

$$\Delta y_i = \delta_0 + \beta_1 \Delta x_i + \Delta u_i$$

What changes:

- Differencing removes a_i exactly
- x_i must vary
- Δu_i uncorrelated with Δx_i
- For usual inference: homoskedastic/no problematic serial dependence in differenced errors

Fixed Effects (FE / Within) Estimator

Within transformation:

$$y_{it} - \bar{y}_i = \beta_1(x_{it} - \bar{x}_i) + (u_{it} - \bar{u}_i)$$

- Removes all time-invariant unobservables a_i
- Strict exogeneity: $\text{Cov}(x_{it}, u_{is}) = 0$ for all t, s
- No need for $\text{Cov}(a_i, x_{it}) = 0$
- Equivalent to FD if only two periods but can be more efficient with $T > 2$

Random Effects (RE) Estimator

RE keeps a_i in composite error but assumes orthogonality:

$$\text{Cov}(a_i, x_{it}) = 0$$

- Bias variance tradeoff: pay the price of this assumption, get higher efficiency

Panel Assumptions

Estimator	Assumption	Solves a_i - x corr.	Time-invariant x
POLS	$\text{Cov}(x_{it}, a_i) = 0$	No	Yes
FD	$\text{Cov}(\Delta x_{it}, \Delta u_{it}) = 0$	Yes	No
FE	$\text{Cov}(x_{it}, u_{is}) = 0 \forall t, s$	Yes	No
RE	$\text{Cov}(x_{it}, a_i) = 0$	No	Yes

Two-period treatment effect representation:

$$\hat{\beta}_{DiD} = \Delta \bar{y}_{treat} - \Delta \bar{y}_{control}$$

Where this fits

Panel methods solve unobserved heterogeneity. Sometimes sacrifice non-time-varying regressors.

- If a_i likely correlated with regressors: FE/FD
- If orthogonality plausible and time-invariant regressors matter: RE

6. Instrumental Variables

Unit 6: The Endogeneity Problem

Structural equation:

$$y = \beta_0 + \beta_1 x + u, \quad \text{Cov}(x, u) \neq 0$$

- Endogeneity (e.g., OVB), biased and inconsistent
- Instrument z :
 - Exogeneity (not testable): $\text{Cov}(z, u) = 0$
 - Relevance (testable): $\text{Cov}(z, x) \neq 0$

$$\hat{\beta} = (Z'X)^{-1}Z'y$$

consistent if z is valid

2SLS Workflow

1st Stage:

$$x = \pi_0 + \pi_1 z + \pi_2 w + v$$

2nd Stage:

$$y = \beta_0 + \beta_1 \hat{x} + \beta_2 w + e$$

Key note:

- Correct 2SLS standard errors are required
- Equivalent to previous one if one instrument, one endogenous

Changes

- Consistent, not efficient
- Weak instruments can create severe finite-sample problems

IV with Multiple Instruments

Benefits:

- More first-stage predictive power can improve precision
- Enables overidentification checks when more instruments than endogenous regressors
- Endogeneity tests (OLS vs IV logic)
- Overidentifying restriction tests (joint instrument exogeneity assumptions)

Takeaway

Where this fits

If endogeneity, IV can solve by “taking exogeneity” from another variable.

7. Simultaneous Equations

Simultaneity as Endogeneity

Example system:

$$y_1 = \alpha_1 y_2 + \beta_1 z_1 + u_1, \quad y_2 = \alpha_2 y_1 + \beta_2 z_2 + u_2$$

What changes:

- Endogenous are jointly determined: regressor in each equation contains part of structural error from the other equation
- OLS inconsistent
- Need excluded exogenous shifters to identify each structural equation
- Order condition (necessary): more excluded exogenous variables than those endogenous (both equations above are identified)
- Rank condition (necessary and sufficient): excluded variables shift endogenous regressor in that equation

Connection to IV

- Each structural equation with endogenous RHS variable can be estimated by 2SLS if identified
- Instruments come from exogenous variables excluded from that equation

Unit 7 Takeaway

Where this fits

SEM formalizes equilibrium/reverse-causality environments and shows why identification design is central before estimation.

Review

Covered

- 1 Stop assuming homoskedasticity
- 2 Explore additional ways to break MLR.4 (endogeneity)
- 3 New data type: time series
- 4 New data type: panel data
- 5 A solution to endogeneity: instrumental variables
- 6 Simultaneous equations